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Applied Calculus IV

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Week 8

- 4.4 (review)
- 4.6

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Higher-Order Differential Equations

- | | |
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| 4.1 | Preliminary Theory—Linear Equations |
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Homogeneous and Nonhomogeneous Linear Differential Equation

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0$$

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = g(x),$$

Recall

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = g(x),$$

$$y = y_c + y_p$$

The general solution of a nonhomogeneous equation = complementary function + any particular solution



general solution of the homogenous version

The key note for finding y_c : **Roots of Characteristic Equation**

For Example: $ay'' + by' + cy = 0$, $\xrightarrow{\text{Define } y = e^{rx}}$ $ar^2 + br + c = 0$

Two distinct real roots r_1 and r_2 $y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$

One repeated real root r $y = (c_1 + c_2 x) e^{rx}$

Complex conjugate roots $r = \alpha \pm \beta i$ $y = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$

The key note for finding y_p : **Undetermined Coefficients**

(Choosing a trial particular solution)

TABLE 4.4.1 Trial Particular Solutions

$g(x)$	Form of y_p
1. 1 (any constant)	A
2. $5x + 7$	$Ax + B$
3. $3x^2 - 2$	$Ax^2 + Bx + C$
4. $x^3 - x + 1$	$Ax^3 + Bx^2 + Cx + E$
5. $\sin 4x$	$A \cos 4x + B \sin 4x$
6. $\cos 4x$	$A \cos 4x + B \sin 4x$
7. e^{5x}	Ae^{5x}
8. $(9x - 2)e^{5x}$	$(Ax + B)e^{5x}$
9. x^2e^{5x}	$(Ax^2 + Bx + C)e^{5x}$
10. $e^{3x} \sin 4x$	$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
11. $5x^2 \sin 4x$	$(Ax^2 + Bx + C) \cos 4x + (Ex^2 + Fx + G) \sin 4x$
12. $xe^{3x} \cos 4x$	$(Ax + B)e^{3x} \cos 4x + (Cx + E)e^{3x} \sin 4x$

Example: Solve $y'' + 4y' - 2y = 2x^2 - 3x + 6$.

Solution

Step1: finding the general solution of homogenous version

Define $y = e^{rx} \rightarrow r^2 + 4r - 2 = 0, \rightarrow$

$$\begin{aligned} r &= \frac{-4 \pm \sqrt{4^2 - 4 \times 1 \times (-2)}}{2 \times 1} \\ &= \frac{-4 \pm \sqrt{16 + 8}}{2} \\ &= \frac{-4 \pm \sqrt{24}}{2} \\ &= \frac{-4 \pm 2\sqrt{6}}{2} \\ &= -2 \pm \sqrt{6} \end{aligned}$$

$e^{(-2+\sqrt{6})x}$ and $e^{(-2-\sqrt{6})x}$ are fundamental solutions

 $y_c = c_1 e^{(-2+\sqrt{6})x} + c_2 e^{(-2-\sqrt{6})x}$

Step2: Finding a particular solution of non-homogenous version

Define $y_p = Ax^2 + Bx + C$. $\longrightarrow y'_p = 2Ax + B$ and $y''_p = 2A$

Replace $y''_p + 4y'_p - 2y_p = 2A + 8Ax + 4B - 2Ax^2 - 2Bx - 2C = 2x^2 - 3x + 6$.

$$\longrightarrow -2A = 2, \quad 8A - 2B = -3, \quad 2A + 4B - 2C = 6.$$

$$\longrightarrow A = -1, B = -\frac{5}{2}, \text{ and } C = -9.$$

$$\longrightarrow y_p = -x^2 - \frac{5}{2}x - 9.$$

Step3: $y = y_c + y_p$

Example: Solve $y'' - y' + y = 2 \sin 3x$

Solution

Step1: finding the general solution of homogenous version

Define $y = e^{rx} \rightarrow r^2 - r + 1 = 0. \rightarrow r = \frac{1 \pm \sqrt{(-1)^2 - 4 \times 1 \times 1}}{2 \times 1}$

$$= \frac{1 \pm \sqrt{1 - 4}}{2}$$
$$= \frac{1 \pm \sqrt{-3}}{2}$$
$$= \frac{1 \pm i\sqrt{3}}{2}$$

$e^{\frac{1}{2}x} \cos \frac{\sqrt{3}}{2}x$ and $e^{\frac{1}{2}x} \sin \frac{\sqrt{3}}{2}x$ are
fundamental solutions

$\rightarrow y_c = e^{\frac{1}{2}x} \left(c_1 \cos \frac{\sqrt{3}}{2}x + c_2 \sin \frac{\sqrt{3}}{2}x \right)$

Step2: Finding a particular solution of non-homogenous version

Define $y_p = A \cos 3x + B \sin 3x.$

→ $y'_p = -3A \sin 3x + 3B \cos 3x$

$$y''_p = -9A \cos 3x - 9B \sin 3x$$

Replace $y''_p - y'_p + y_p = (-8A - 3B) \cos 3x + (3A - 8B) \sin 3x = 2 \sin 3x$

→ $-8A - 3B = 0, \quad 3A - 8B = 2, \quad \rightarrow A = \frac{6}{73} \text{ and } B = -\frac{16}{73}$

→ $y_p = \frac{6}{73} \cos 3x - \frac{16}{73} \sin 3x.$

Step3: $y = y_c + y_p$

Example: Solve $y'' - y' + y = 5e^{2x}$.

Solution

Step1: finding the general solution of homogenous version

Define $y = e^{rx} \rightarrow r^2 - r + 1 = 0. \rightarrow r = \frac{1 \pm \sqrt{(-1)^2 - 4 \times 1 \times 1}}{2 \times 1}$

$$= \frac{1 \pm \sqrt{1 - 4}}{2}$$
$$= \frac{1 \pm \sqrt{-3}}{2}$$
$$= \frac{1 \pm i\sqrt{3}}{2}$$

$e^{\frac{1}{2}x} \cos \frac{\sqrt{3}}{2}x$ and $e^{\frac{1}{2}x} \sin \frac{\sqrt{3}}{2}x$ are
fundamental solutions

$\rightarrow y_c = e^{\frac{1}{2}x} \left(c_1 \cos \frac{\sqrt{3}}{2}x + c_2 \sin \frac{\sqrt{3}}{2}x \right)$

Step2: Finding a particular solution of non-homogenous version

Define $y_p = Ae^{2x}$

$\longrightarrow y'_p = 2Ae^{2x}$

$$y''_p = 4Ae^{2x}$$

Replace in $y'' - y' + y = 5e^{2x}$

$$\begin{aligned} y''_p - y'_p + y_p &= 4Ae^{2x} - 2Ae^{2x} + Ae^{2x} \\ &= (4A - 2A + A)e^{2x} \\ &= 3Ae^{2x} \end{aligned}$$

$$\longrightarrow A = \frac{5}{3} \quad \longrightarrow y_p = \frac{5}{3}e^{2x}$$

Step3: $y = y_c + y_p$

Example: Solve $y'' - 2y' - 3y = 4x - 5 + 6xe^{2x}$

Solution

Step1: finding the general solution of homogenous version

Define $y = e^{rx} \rightarrow r^2 - 2r - 3 = 0 \rightarrow (r - 3)(r + 1) = 0$

e^{-x} and e^{3x} are fundamental solutions

$\rightarrow y_c = c_1e^{-x} + c_2e^{3x}$

Step2: Finding a particular solution of non-homogenous version

Define $y_p = Ax + B + Cxe^{2x} + Ee^{2x}$

→ $y'_p = A + Ce^{2x} + 2Cxe^{2x} + 2Ee^{2x}$

$$y''_p = 4Ce^{2x} + 4Cxe^{2x} + 4Ee^{2x}$$

Replace in $y'' - 2y' - 3y = 4x - 5 + 6xe^{2x}$

$$-3Ax - 2A - 3B - 3Cxe^{2x} + (2C - 3E)e^{2x} = 4x - 5 + 6xe^{2x}$$

→ $-3A = 4, \quad -2A - 3B = -5, \quad -3C = 6, \quad 2C - 3E = 0.$

→ $y_p = -\frac{4}{3}x + \frac{23}{9} - 2xe^{2x} - \frac{4}{3}e^{2x}$

$$A = -\frac{4}{3}$$

$$B = \frac{23}{9}$$

$$C = -2$$

$$E = -\frac{4}{3}$$

Step3: $y = y_c + y_p$

Remark: Using the Superposition Principle

$$y'' - 9y' + 14y = 3x^2 - 5 \sin 2x + 8xe^{6x} \quad \rightarrow$$

$$y_p = y_{p_1} + y_{p_2} + y_{p_3}$$

$$y_{p_1} = Ax^2 + Bx + C,$$

$$y_{p_2} = E \cos 2x + F \sin 2x,$$

$$y_{p_3} = Gxe^{6x} + He^{6x}.$$

break it to three parts and find their particular solutions separately

$$y'' - 9y' + 14y = 3x^2$$

$$y'' - 9y' + 14y = -5 \sin 2x$$

$$y'' - 9y' + 14y = 8xe^{6x}$$

✓ We can find these particular solutions separately or simultaneously

Remark: No overlap between y_p and y_c

To define the particular solution y_p of a linear nonhomogeneous ODE:

- ✓ Identify all fundamental solutions of homogenous version (basis terms in the complementary solution y_c)
- ✓ If your trial y_p shares any term with y_c , multiply the trial by the smallest power of x to eliminate the match
(Avoid overlap).

Because:

The basis terms in y_c are the solutions of homogenous version and vanish the left side, and won't satisfy the nonhomogenous version

Example: Solve $y'' - 2y' - 3y = 4e^{3x}$

Solution

Step1: finding the general solution of homogenous version

Define $y = e^{rx} \rightarrow r^2 - 2r - 3 = 0 \rightarrow (r - 3)(r + 1) = 0$

e^{-x} and e^{3x} are fundamental solutions

$\rightarrow y_c = c_1 e^{-x} + c_2 e^{3x}$

Define $y_p = Ae^{3x}$

$$\longrightarrow y'_p = 3Ae^{3x}$$

$$y''_p = 9Ae^{3x}$$

Replace in $y'' - 2y' - 3y = 4e^{3x}$

$$9Ae^{3x} - 2(3Ae^{3x}) - 3(Ae^{3x}) = 4e^{3x}$$

$= 0$

$\longrightarrow$ It doesn't give any solution

$y = Ae^{3x}$ can't be a solution of nonhomogenous version

Step2: Finding a particular solution of non-homogenous version

Define

$$y_p = Axe^{3x}$$



$$y_p' = Ae^{3x} + 3Axe^{3x}$$

$$y_p'' = 3Ae^{3x} + 3Ae^{3x} + 9Axe^{3x} = 6Ae^{3x} + 9Axe^{3x}$$

Replace in

$$y'' - y' + y = 5e^{2x}$$

$$(6Ae^{3x} + 9Axe^{3x}) - 2(Ae^{3x} + 3Axe^{3x}) - 3(Axe^{3x}) = 4e^{3x}$$

$$6Ae^{3x} + 9Axe^{3x} - 2Ae^{3x} - 6Axe^{3x} - 3Axe^{3x} = 4e^{3x}$$

$$4Ae^{3x} = 4e^{3x} \quad \longrightarrow \quad A = 1$$

$$\longrightarrow y_p = xe^{3x}$$

$$\text{Step3: } y = y_c + y_p$$

Example: Solve $y^{(4)} + y''' = 1 - x^2 e^{-x}$

Solution

Step1: finding the general solution of homogenous version

Define $y = e^{rx} \rightarrow r^4 + r^3 = 0 \rightarrow r^3(r + 1) = 0$

$e^{0x}, xe^{0x}, x^2e^{0x}$ and e^{-x} are fundamental solutions

 $y_c = c_1 + c_2x + c_3x^2 + c_4e^{-x}$

Step2: Finding a particular solution of non-homogenous version

$$g(x) = 1 - x^2 e^{-x}$$

→ $y_p = A + (Bx^2 + Cx + D)e^{-x}$ However we may initiate with this guess,
since the constant and e^{-x} are seen in y_c

$$y_c = c_1 + c_2 x + c_3 x^2 + c_4 e^{-x}$$

so we change the trial y_p

Define $y_p = Ax^3 + x(Bx^2 + Cx + D)e^{-x}$

Exercises 4.4

In Problems 1–26 solve the given differential equation by undetermined coefficients.

1. $y'' + 3y' + 2y = 6$

2. $4y'' + 9y = 15$

3. $y'' - 10y' + 25y = 30x + 3$

4. $y'' + y' - 6y = 2x$

5. $\frac{1}{4}y'' + y' + y = x^2 - 2x$

6. $y'' - 8y' + 20y = 100x^2 - 26xe^x$

7. $y'' + 3y = -48x^2e^{3x}$

8. $4y'' - 4y' - 3y = \cos 2x$

9. $y'' - y' = -3$

10. $y'' + 2y' = 2x + 5 - e^{-2x}$

11. $y'' - y' + \frac{1}{4}y = 3 + e^{x/2}$

12. $y'' - 16y = 2e^{4x}$

13. $y'' + 4y = 3 \sin 2x$

14. $y'' - 4y = (x^2 - 3) \sin 2x$

15. $y'' + y = 2x \sin x$

16. $y'' - 5y' = 2x^3 - 4x^2 - x + 6$

17. $y'' - 2y' + 5y = e^x \cos 2x$

Solution 1. $y'' + 3y' + 2y = 6$

Step 1. general solution of the homogeneous equation:

Define $y = e^{rx} \rightarrow r^2 + 3r + 2 = 0$ (characteristic equation)

$$\rightarrow \text{roots } r_1 = -1, r_2 = -2 \rightarrow y_c = c_1 e^{-x} + c_2 e^{-2x}$$

Step2. particular solution

Define $y_p = A \rightarrow$ substitute into the the non-homogeneous equation

$$2A = 6 \rightarrow A = 3 \rightarrow y_p = 3$$

Step 3. general solution of the original equation:

$$y = y_c + y_p = c_1 e^{-x} + c_2 e^{-2x} + 3$$

Solution 2. $4y'' + 9y = 15$

Step 1. general solution of the homogeneous equation:

Define $y = e^{rx} \rightarrow 4r^2 + 9 = 0$ (characteristic equation)

$$\rightarrow \text{roots } r_1, r_2 = \pm \frac{3}{2}i \rightarrow y_c = c_1 e^{0x} \cos \frac{3}{2}x + c_2 e^{0x} \sin \frac{3}{2}x = c_1 \cos \frac{3}{2}x + c_2 \sin \frac{3}{2}x$$

Step2. particular solution

Define $y_p = A \rightarrow$ substitute into the the non-homogeneous equation

$$9A = 15 \rightarrow A = \frac{5}{3} \rightarrow y_p = \frac{5}{3}$$

Step 3. general solution of the original equation:

$$y = y_c + y_p = c_1 \cos \frac{3}{2}x + c_2 \sin \frac{3}{2}x + \frac{5}{3}$$

Solution 3. $y'' - 10y' + 25y = 30x + 3$

Step 1. General solution of the homogeneous equation:

Define $y = e^{rx} \rightarrow r^2 - 10r + 25 = 0$ (characteristic equation)

$\rightarrow$ roots $r_1, r_2 = 5$ (repeated) $\rightarrow y_c = c_1 e^{5x} + c_2 x e^{5x}$

Step2. particular solution

Define $y_p = Ax + B \rightarrow$ substitute into the the non-homogeneous equation

$$-10A + 25Ax + 25B = 30x + 3$$

$$\rightarrow 25A = 30, -10A + 25B = 3 \rightarrow A = \frac{6}{5}, B = \frac{3}{5} \rightarrow y_p = \frac{6}{5}x + \frac{3}{5}$$

Step 3. general solution of the original equation:

$$y = y_c + y_p = c_1 e^{5x} + c_2 x e^{5x} + \frac{6}{5}x + \frac{3}{5}$$

Solution 10. $y'' + 2y' = 2x + 5 - e^{-2x}$

Step 1. General solution of the homogeneous equation:

Define $y = e^{rx} \rightarrow r^2 + 2r = 0$ (characteristic equation) $\rightarrow$ roots $r_1 = 0, r_2 = -2$

$$\rightarrow y_c = c_1 + c_2 e^{-2x}$$

Step 2. Particular solution

(Note that $y_p = Ax + B + Ce^{-2x}$ might be your first guess but it has overlap with y_c)

Define

$$\begin{aligned} y_p &= Ax^2 + Bx + Cxe^{-2x} \\ y_p' &= 2Ax + B + Ce^{-2x} - 2Cxe^{-2x} \\ y_p'' &= 2A - 4Ce^{-2x} + 4Cxe^{-2x} \end{aligned}$$

Substitution $4Ax + 2A + 2B - 2Ce^{-2x} = 2x + 5 - e^{-2x}$

$$\rightarrow 4A = 2, 2A + 2B = 5, -2C = -1 \rightarrow A = \frac{1}{2}, B = 2, C = \frac{1}{2} \rightarrow y_p = \frac{1}{2}x^2 + 2x + \frac{1}{2}xe^{-2x}$$

Step 3. General solution of the original equation:

$$y = y_c + y_p = c_1 + c_2 e^{-2x} + \frac{1}{2}x^2 + 2x + \frac{1}{2}x e^{-2x}$$

Solution 13. $y'' + 4y = 3 \sin 2x$

Step 1. General solution of the homogeneous equation:

Define $y = e^{rx} \rightarrow r^2 + 4r = 0$ (characteristic equation) $\rightarrow$ roots $r_1, r_2 = \pm 2i$

$$\rightarrow y_c = c_1 \cos 2x + c_2 \sin 2x$$

Step 2. Particular solution

(Note that $y_p = A \cos 2x + B \sin 2x$ might be your first guess but it has overlap with y_c)

Define

$$y_p = Ax \cos 2x + Bx \sin 2x$$

$$y_p' = A \cos 2x - 2Ax \sin 2x + B \sin 2x + 2Bx \cos 2x$$

$$y_p'' = -4A \sin 2x - 4Ax \cos 2x + 4B \cos 2x - 4Bx \sin 2x$$

Substitution $-4A \sin 2x + 4B \cos 2x = 3 \sin 2x$

$$\rightarrow -4A = 3, 4B = 0, \rightarrow A = -\frac{3}{4}, B = 0 \rightarrow y_p = -\frac{3}{4}x \cos 2x$$

Step 3. General solution of the original equation:

$$y = y_c + y_p = c_1 \cos 2x + c_2 \sin 2x - \frac{3}{4}x \cos 2x$$

Solution 13. $y'' + y = 2x \sin x$

Step 1. General solution of the homogeneous equation:

Define $y = e^{rx} \rightarrow r^2 + r = 0$ (characteristic equation) $\rightarrow$ roots $r_1, r_2 = \pm i$

$$\rightarrow y_c = c_1 \cos x + c_2 \sin x$$

Step 2. Particular solution

(Note that $y_p = (Ax + B) \cos x + (Cx + D) \sin x$ might be your first guess but it has overlap with y_c)

Define $y_p = (Ax^2 + Bx) \cos x + (Cx^2 + Dx) \sin x$

Substitution :

$$y''_p + y'_p = (4Cx + 2A + 2D) \cos x + (-4Ax + 2C - 2B) \sin x = 2x \sin x$$

$$\rightarrow -4A = 2, 4C = 0, 2A + 2D = 0, 2C - 2B = 0 \rightarrow A = -\frac{1}{2}, B = C = 0, D = \frac{1}{2}$$

$$\rightarrow y_p = -\frac{1}{2} x^2 \cos x + \frac{1}{2} x \sin x$$

Step 3. General solution of the original equation:

$$y = y_c + y_p = c_1 \cos x + c_2 \sin x - \frac{1}{2} x^2 \cos x + \frac{1}{2} x \sin x$$

18. $y'' - 2y' + 2y = e^{2x}(\cos x - 3 \sin x)$

19. $y'' + 2y' + y = \sin x + 3 \cos 2x$

20. $y'' + 2y' - 24y = 16 - (x + 2)e^{4x}$

21. $y''' - 6y'' = 3 - \cos x$

22. $y''' - 2y'' - 4y' + 8y = 6xe^{2x}$

23. $y''' - 3y'' + 3y' - y = x - 4e^x$

24. $y''' - y'' - 4y' + 4y = 5 - e^x + e^{2x}$

25. $y^{(4)} + 2y'' + y = (x - 1)^2$

26. $y^{(4)} - y'' = 4x + 2xe^{-x}$

Solution 19. $y'' + 2y' + y = \sin x + 3 \cos 2x$

Step 1. General solution of the homogeneous equation:

Define $y = e^{rx} \rightarrow r^2 + 2r + r = 0$ (characteristic equation) $\rightarrow$ roots $r_1, r_2 = -1$ (repeated)

$$\rightarrow y_c = c_1 e^{-x} + c_2 x e^{-x}$$

Step2. Particular solution

Define $y_p = A \cos x + B \sin x + C \cos 2x + Cx \sin 2x$

Substitute in the original equation

$$A = -\frac{1}{2}, \quad B = 0, \quad C = -\frac{9}{25}, \quad D = \frac{12}{25} \rightarrow y_p = -\frac{1}{2} \cos x - \frac{9}{25} \cos 2x + \frac{12}{25} x \sin 2x$$

Step2.

$$y = y_c + y_p = c_1 e^{-x} + c_2 x e^{-x} + -\frac{1}{2} \cos x - \frac{9}{25} \cos 2x + \frac{12}{25} x \sin 2x$$

Solution 25. $y^{(4)} + 2y'' + y = (x - 1)^2$

Step 1. General solution of the homogeneous equation:

Define $y = e^{rx} \rightarrow r^4 + 2r^2 + r = 0$ (characteristic equation) $\rightarrow$ roots $r_1, r_2 = \pm i$ (repeated)

$$\rightarrow y_c = c_1 \cos x + c_2 \sin x + c_3 x \cos x + c_4 x \sin x$$

Step2. Particular solution

Define $y_p = Ax^2 + Bx + C$

Substitution $0 + 4A + Ax^2 + Bx + C = x^2 - 2x + 1,$

$$A = 1, \quad B = -2, \quad 4A + C = 1 \rightarrow y_p = x^2 - 2x - 3$$

Step2.

$$y = y_c + y_p = c_1 \cos x + c_2 \sin x + c_3 x \cos x + c_4 x \sin x + x^2 - 2x - 3$$

In Problems 27–36 solve the given initial-value problem.

27. $y'' + 4y = -2$, $y(\pi/8) = \frac{1}{2}$, $y'(\pi/8) = 2$

28. $2y'' + 3y' - 2y = 14x^2 - 4x - 11$, $y(0) = 0$, $y'(0) = 0$

29. $5y'' + y' = -6x$, $y(0) = 0$, $y'(0) = -10$

30. $y'' + 4y' + 4y = (3 + x)e^{-2x}$, $y(0) = 2$, $y'(0) = 5$

31. $y'' + 4y' + 5y = 35e^{-4x}$, $y(0) = -3$, $y'(0) = 1$

32. $y'' - y = \cosh x$, $y(0) = 2$, $y'(0) = 12$

33. $\frac{d^2x}{dt^2} + \omega^2x = F_0 \sin \omega t$, $x(0) = 0$, $x'(0) = 0$

34. $\frac{d^2x}{dt^2} + \omega^2x = F_0 \cos \gamma t$, $x(0) = 0$, $x'(0) = 0$

35. $y''' - 2y'' + y' = 2 - 24e^x + 40e^{5x}$, $y(0) = \frac{1}{2}$, $y'(0) = \frac{5}{2}$,
 $y''(0) = -\frac{9}{2}$

36. $y''' + 8y = 2x - 5 + 8e^{-2x}$, $y(0) = -5$, $y'(0) = 3$,
 $y''(0) = -4$

Solution 27. $y'' + 4y = -2, \quad y(\pi/8) = \frac{1}{2}, y'(\pi/8) = 2$

Find the general solution of this equation:

$$y = y_c + y_p = c_1 \cos 2x + c_2 \sin 2x - \frac{1}{2}$$

Apply the conditions

$$y\left(\frac{\pi}{8}\right) = c_1 \cos \frac{\pi}{4} + c_2 \sin \frac{\pi}{4} - \frac{1}{2} = \frac{1}{2}$$

$$y'\left(\frac{\pi}{8}\right) = -2c_1 \sin \frac{\pi}{4} + 2c_2 \cos \frac{\pi}{4} = 2$$

$$\rightarrow c_1 = 0, c_2 = \sqrt{2}$$

$$\rightarrow y = \sqrt{2} \sin 2x - \frac{1}{2}$$

In Problems 37–40 solve the given boundary-value problem.

37. $y'' + y = x^2 + 1, \quad y(0) = 5, y(1) = 0$

38. $y'' - 2y' + 2y = 2x - 2, \quad y(0) = 0, y(\pi) = \pi$

39. $y'' + 3y = 6x, \quad y(0) = 0, y(1) + y'(1) = 0$

40. $y'' + 3y = 6x, \quad y(0) + y'(0) = 0, y(1) = 0$

Challenge

We want to find a particular solution for other $g(x)$ such as

$$g(x) = \ln x, \quad g(x) = \frac{1}{x}, \quad g(x) = \tan x, \quad g(x) = \sin^{-1}x,$$

4.6 Variation of Parameters

(A technique for finding a particular solution)

$$y'' + Py' + Qy = f(x)$$

General solution of the homogeneous equation:

$$\rightarrow y_c = c_1 y_1 + c_2 y_2$$

Now we introduce a trial particular solution as $\rightarrow y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$

(u_1 and u_2 are functions)

$$y_p = u_1 y_1 + u_2 y_2$$

$$y'_p = u'_1 y_1 + u_1 y'_1 + u'_2 y_2 + u_2 y'_2$$

$$y''_p = u''_1 y_1 + 2u'_1 y'_1 + u_1 y''_1 + u''_2 y_2 + 2u'_2 y'_2 + u_2 y''_2$$

Replace them in the equation,
simplify and rearrange the
terms

$$\begin{aligned}
 y'' + Py' + Qy &= u_1 \overbrace{[y''_1 + Py'_1 + Qy_1]}^{\text{zero}} + u_2 \overbrace{[y''_2 + Py'_2 + Qy_2]}^{\text{zero}} + y_1 u''_1 + u'_1 y'_1 \\
 &\quad + y_2 u''_2 + u'_2 y'_2 + P[y_1 u'_1 + y_2 u'_2] + y'_1 u'_1 + y'_2 u'_2 \\
 &= \frac{d}{dx} [y_1 u'_1] + \frac{d}{dx} [y_2 u'_2] + P[y_1 u'_1 + y_2 u'_2] + y'_1 u'_1 + y'_2 u'_2 \\
 &= \frac{d}{dx} [y_1 u'_1 + y_2 u'_2] + P[y_1 u'_1 + y_2 u'_2] + y'_1 u'_1 + y'_2 u'_2 = f(x).
 \end{aligned}$$

If we assume

$$u'_1 y_1 + u'_2 y_2 = 0$$

$$\frac{d}{dx} [y_1 u'_1 + y_2 u'_2] + P[y_1 u'_1 + y_2 u'_2] + y'_1 u'_1 + y'_2 u'_2 = f(x).$$

$\underbrace{\hspace{10em}}_{=0}$ $\underbrace{\hspace{10em}}_{=0}$

$$\Rightarrow \begin{cases} u'_1 y_1 + u'_2 y_2 = 0 \\ u'_1 y'_1 + u'_2 y'_2 = f(x) \end{cases}$$

$$u'_1 = \frac{W_1}{W} = -\frac{y_2 f(x)}{W} \quad \text{and} \quad u'_2 = \frac{W_2}{W} = \frac{y_1 f(x)}{W},$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y'_1 & y'_2 \end{vmatrix}, \quad W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y'_2 \end{vmatrix}, \quad W_2 = \begin{vmatrix} y_1 & 0 \\ y'_1 & f(x) \end{vmatrix}.$$

Recall: Cramer's Method

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases} \rightarrow x = \frac{D_x}{D}, \quad y = \frac{D_y}{D}$$

where

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = a_1b_2 - a_2b_1$$

$$D_x = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix} = c_1b_2 - c_2b_1$$

$$D_y = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix} = a_1c_2 - a_2c_1$$

Example: Solve $y'' - 4y' + 4y = (x + 1)e^{2x}$.

Solution

General solution of the homogeneous equation:

Define $y = e^{rx} \rightarrow r^2 + 2r = 0$ (characteristic equation) $\rightarrow$ roots $r_1, r_2 = 2$ (repeated)

$\rightarrow e^{2x}$ and xe^{2x} are fundamental solutions

$\rightarrow y_1 = e^{2x}$ and $y_2 = xe^{2x} \rightarrow y_p = u_1 e^{2x} + u_2 xe^{2x}$

$$W_1 = \begin{vmatrix} 0 & xe^{2x} \\ (x+1)e^{2x} & 2xe^{2x} + e^{2x} \end{vmatrix} = -(x+1)xe^{4x}, \quad W_2 = \begin{vmatrix} e^{2x} & 0 \\ 2e^{2x} & (x+1)e^{2x} \end{vmatrix} = (x+1)e^{4x},$$

$$W(e^{2x}, xe^{2x}) = \begin{vmatrix} e^{2x} & xe^{2x} \\ 2e^{2x} & 2xe^{2x} + e^{2x} \end{vmatrix} = e^{4x}.$$

➡ $u_1' = -\frac{(x+1)xe^{4x}}{e^{4x}} = -x^2 - x, \quad u_2' = \frac{(x+1)e^{4x}}{e^{4x}} = x + 1.$

➡ $u_1 = -\frac{1}{3}x^3 - \frac{1}{2}x^2 \quad \text{and} \quad u_2 = \frac{1}{2}x^2 + x.$

➡ $y_p = \left(-\frac{1}{3}x^3 - \frac{1}{2}x^2\right)e^{2x} + \left(\frac{1}{2}x^2 + x\right)xe^{2x} = \frac{1}{6}x^3e^{2x} + \frac{1}{2}x^2e^{2x}$

Finally, $y = y_c + y_p = c_1 e^{2x} + c_2 x e^{2x} + \frac{1}{6} x^3 e^{2x} + \frac{1}{2} x^2 e^{2x}.$

Example: Solve $4y'' + 36y = \csc 3x$.

Solution

Step 1. General solution of the homogeneous equation:

Define $y = e^{rx} \rightarrow 4r^2 + 36 = 0$ (characteristic equation) $\rightarrow$ roots $r_1, r_2 = \pm 3i$

$\rightarrow \cos 3x$ and $\sin 3x$ are fundamental solutions

Step 2. $\rightarrow y_1 = \cos 3x$ and $y_2 = \sin 3x \rightarrow y_p = u_1 \cos 3x + u_2 \sin 3x$

Remark: The formulas

$$u'_1 = \frac{W_1}{W} = \frac{\begin{vmatrix} 0 & y_2 \\ f(x) & y'_2 \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y'_1 & y'_2 \end{vmatrix}}$$

$$u'_2 = \frac{W_2}{W} = \frac{\begin{vmatrix} y_1 & 0 \\ y'_1 & f(x) \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y'_1 & y'_2 \end{vmatrix}}$$

are valid for this version,

$$y'' + Py' + Qy = f(x)$$

So, in this example, to use the formulas, we first divide the equation by 4, $y'' + 9y = \frac{1}{4} \csc 3x$

$$y'' + 9y = \overset{f(x)}{\frac{1}{4} \csc 3x}$$

$$W(\cos 3x, \sin 3x) = \begin{vmatrix} \cos 3x & \sin 3x \\ -3 \sin 3x & 3 \cos 3x \end{vmatrix} = 3,$$

$$W_1 = \begin{vmatrix} 0 & \sin 3x \\ \frac{1}{4} \csc 3x & 3 \cos 3x \end{vmatrix} = -\frac{1}{4}, \quad W_2 = \begin{vmatrix} \cos 3x & 0 \\ -3 \sin 3x & \frac{1}{4} \csc 3x \end{vmatrix} = \frac{1}{4} \frac{\cos 3x}{\sin 3x}.$$

➡ $u'_1 = \frac{W_1}{W} = -\frac{1}{12} \quad \text{and} \quad u'_2 = \frac{W_2}{W} = \frac{1}{12} \frac{\cos 3x}{\sin 3x}$

➡ $u_1 = -\frac{1}{12}x \quad \text{and} \quad u_2 = \frac{1}{36} \ln |\sin 3x|.$

➡ $y_p = -\frac{1}{12}x \cos 3x + \frac{1}{36} (\sin 3x) \ln |\sin 3x|$

Finally,

$$y = y_c + y_p = c_1 \cos 3x + c_2 \sin 3x - \frac{1}{12} x \cos 3x + \frac{1}{36} (\sin 3x) \ln |\sin 3x|.$$

Exercises 4.6

In Problems 1–18 solve each differential equation by variation of parameters.

1. $y'' + y = \sec x$

3. $y'' + y = \sin x$

5. $y'' + y = \cos^2 x$

7. $y'' - y = \cosh x$

9. $y'' - 9y = \frac{9x}{e^{3x}}$

11. $y'' + 3y' + 2y = \frac{1}{1 + e^x}$

12. $y'' - 2y' + y = \frac{e^x}{1 + x^2}$

13. $y'' + 3y' + 2y = \sin e^x$

14. $y'' - 2y' + y = e^t \arctan t$

15. $y'' + 2y' + y = e^{-t} \ln t$

16. $2y'' + y' = 6x$

17. $3y'' - 6y' + 6y = e^x \sec x$

18. $4y'' - 4y' + y = e^{x/2} \sqrt{1 - x^2}$

2. $y'' + y = \tan x$

4. $y'' + y = \sec \theta \tan \theta$

6. $y'' + y = \sec^2 x$

8. $y'' - y = \sinh 2x$

10. $4y'' - y = e^{x/2} + 3$

Solution 1 $y'' + y = \sec x$

Solution

Step1. General solution of the homogeneous equation:

Define $y = e^{rx} \rightarrow r^2 + 1 = 0$ (characteristic equation) $\rightarrow$ roots $r_1, r_2 = \pm i$

$\rightarrow \cos x$ and $\sin x$ are fundamental solutions $\rightarrow y_c = c_1 \cos x + c_2 \sin x$

Step2. Particular solution $y_1 = \cos x$ and $y_2 = \sin x \rightarrow y_p = u_1 \cos x + u_2 \sin x$

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1$$

$$W_1 = \begin{vmatrix} 0 & \sin x \\ \sec x & \cos x \end{vmatrix} = 0 \cdot \cos x - \sec x \cdot \sin x = -\frac{\sin x}{\cos x} = -\tan x$$

$$W_2 = \begin{vmatrix} \cos x & 0 \\ -\sin x & \sec x \end{vmatrix} = \cos x \cdot \sec x - 0$$



$$u_1' = \frac{W_1}{W} = \frac{-\tan x}{1} = -\tan x$$

$$u_2' = \frac{W_2}{W} = \frac{1}{1} = 1$$



$$u_1 = \int -\tan x \, dx = \ln |\cos x|$$

$$u_2 = \int 1 \, dx = x$$

$$\rightarrow y_p = \ln|\cos x| \cos x + x \sin x$$

Step3. $y = y_c + y_p = c_1 \cos x + c_2 \sin x + \ln|\cos x| \cos x + x \sin x$

Solution 11 $y'' + 3y' + 2y = \frac{1}{1 + e^x}$

Solution

Step1. General solution of the homogeneous equation:

Define $y = e^{rx} \rightarrow r^2 + 3r + 2 = 0$ (characteristic equation) $\rightarrow$ roots $r_1 = -1, r_2 = -2$

$\rightarrow e^{-x}$ and e^{-2x} are fundamental solutions $\rightarrow y_c = c_1 e^{-x} + c_2 e^{-2x}$

Step2. Particular solution $\rightarrow y_1 = e^{2x}$ and $y_2 = xe^{2x} \rightarrow y_p = u_1 e^{-x} + u_2 e^{-2x}$

$$W = \begin{vmatrix} e^{-x} & e^{-2x} \\ -e^{-x} & -2e^{-2x} \end{vmatrix} = -e^{-3x}$$

$$W_1 = \begin{vmatrix} 0 & e^{-2x} \\ \frac{1}{1+e^x} & -2e^{-2x} \end{vmatrix} = -\frac{e^{-2x}}{1+e^x}$$

$$W_2 = \begin{vmatrix} e^{-x} & 0 \\ -e^{-x} & \frac{1}{1+e^x} \end{vmatrix} = \frac{e^{-x}}{1+e^x}$$



$$u'_1 = \frac{W_1}{W} = \frac{-\frac{e^{-2x}}{1+e^x}}{-e^{-3x}} = \frac{e^x}{1+e^x}$$

$$u'_2 = \frac{W_2}{W} = \frac{\frac{e^{-x}}{1+e^x}}{-e^{-3x}} = -\frac{e^{2x}}{1+e^x}$$

$$\longrightarrow u_1 = \int \frac{e^x}{1+e^x} dx = \ln(1 + e^x) + c_1$$

$$\longrightarrow u_2 = \int -\frac{e^{2x}}{1+e^x} dx = -e^x + \ln(1 + e^x) + c_2$$

Remark: we usually, assume $c_1 = c_2 = 0$,
because we only need one particular solution

$$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$$



$$y_p = e^{-x} \ln(1 + e^x) - e^{-x} + e^{-2x} \ln(1 + e^x)$$

Step3. $y = y_c + y_p$

$$y = c_1 e^{-x} + c_2 e^{-2x} + e^{-x} \ln(1 + e^x) - e^{-x} + e^{-2x} \ln(1 + e^x)$$

$$y = (c_1 - 1)e^{-x} + c_2 e^{-2x} + e^{-x} \ln(1 + e^x) + e^{-2x} \ln(1 + e^x)$$

$$y = c_1 e^{-x} + c_2 e^{-2x} + e^{-x} \ln(1 + e^x) + e^{-2x} \ln(1 + e^x)$$

In Problems 19–22 solve each differential equation by variation of parameters, subject to the initial conditions $y(0) = 1$, $y'(0) = 0$.

19. $4y'' - y = xe^{x/2}$

20. $2y'' + y' - y = x + 1$

21. $y'' + 2y' - 8y = 2e^{-2x} - e^{-x}$

22. $y'' - 4y' + 4y = (12x^2 - 6x)e^{2x}$

Solution 21 $y'' + 2y' - 8y = 2e^{-2x} - e^{-x}$

$$y(0) = 1, y'(0) = 0$$

Solution

Step1. General solution of the homogeneous equation:

Define $y = e^{rx} \rightarrow r^2 + 2r - 8 = 0$ (characteristic equation) $\rightarrow$ roots $r_1 = 2, r_2 = -4$

$\rightarrow e^{2x}$ and e^{-4x} are fundamental solutions $\rightarrow y_c = c_1 e^{2x} + c_2 e^{-4x}$

Step2. Particular solution $\rightarrow y_1 = e^{2x}$ and $y_2 = e^{-4x} \rightarrow y_p = u_1 e^{2x} + u_2 e^{-4x}$

$$W = \begin{vmatrix} e^{2x} & e^{-4x} \\ 2e^{2x} & -4e^{-4x} \end{vmatrix} = e^{2x}(-4e^{-4x}) - e^{-4x}(2e^{2x}) = -4e^{-2x} - 2e^{-2x} = -6e^{-2x}$$

$$W_1 = \begin{vmatrix} 0 & e^{-4x} \\ 2e^{-2x} - e^{-x} & -4e^{-4x} \end{vmatrix} = 0(-4e^{-4x}) - e^{-4x}(2e^{-2x} - e^{-x}) = -2e^{-6x} + e^{-5x}$$

$$W_2 = \begin{vmatrix} e^{2x} & 0 \\ 2e^{2x} & 2e^{-2x} - e^{-x} \end{vmatrix} = e^{2x}(2e^{-2x} - e^{-x}) - 0(2e^{2x}) = 2 - e^x$$



$$u'_1 = \frac{W_1}{W} = \frac{-2e^{-6x} + e^{-5x}}{-6e^{-2x}} = \frac{2e^{-4x} - e^{-3x}}{6}$$

$$u'_2 = \frac{W_2}{W} = \frac{2 - e^x}{-6e^{-2x}} = \frac{e^x - 2}{6} e^{2x} = \frac{e^{3x} - 2e^{2x}}{6}$$

$$u_1 = \int \frac{2e^{-4x} - e^{-3x}}{6} dx = -\frac{e^{-4x}}{12} + \frac{e^{-3x}}{18}$$

$$u_2 = \int \frac{e^{3x} - 2e^{2x}}{6} dx = \frac{e^{3x}}{18} - \frac{e^{2x}}{6}$$



$$y_p = \left(-\frac{e^{-4x}}{12} + \frac{e^{-3x}}{18} \right) e^{2x} + \left(\frac{e^{3x}}{18} - \frac{e^{2x}}{6} \right) e^{-4x}$$

$$y_p = -\frac{e^{-2x}}{12} + \frac{e^{-x}}{18} + \frac{e^{-x}}{18} - \frac{e^{-2x}}{6}$$

$$y_p = -\frac{e^{-2x}}{4} + \frac{e^{-x}}{9}$$

Step3. $y = y_c + y_p = c_1 e^{2x} + c_2 e^{-4x} - \frac{1}{4} e^{-2x} + \frac{1}{9} e^{-x}$

Step 4. Apply the conditions and find parameters

$$y = c_1 e^{2x} + c_2 e^{-4x} - \frac{e^{-2x}}{4} + \frac{e^{-x}}{9}$$

$$y(0) = 1 \implies 1 = c_1 + c_2 - \frac{1}{4} + \frac{1}{9} \implies c_1 + c_2 = \frac{41}{36}$$

$$y' = 2c_1 e^{2x} - 4c_2 e^{-4x} + \frac{e^{-2x}}{2} - \frac{e^{-x}}{9}$$

$$y'(0) = 0 \implies 0 = 2c_1 - 4c_2 + \frac{1}{2} - \frac{1}{9} \implies c_1 - 2c_2 = -\frac{7}{36}$$

$$c_1 = \frac{25}{36}, c_2 = \frac{4}{9}$$



$$y = \frac{25}{36} e^{2x} + \frac{4}{9} e^{-4x} - \frac{e^{-2x}}{4} + \frac{e^{-x}}{9}$$